

Thm. Let $\{a_n\}$ be a sequence of Complex Numbers.

If the series $\sum_{n=1}^{\infty} a_n$ Converges absolutely, then for any permutation $\sigma: \mathbb{N} \rightarrow \mathbb{N}$,
the rearrangement $\sum_{n=1}^{\infty} a_{\sigma(n)}$ Converges ^{the} Same value.

Proof Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $m > n \geq N$ implies $\sum_{k=n}^m |a_k| < \varepsilon$
Take $S_i = \sum_{i=1}^i a_i$ and $S'_i = \sum_{i=1}^i a_{\sigma(i)}$.

Take $M \in \mathbb{N}$ s.t. $\{1, 2, \dots, N-1\} \subseteq \{\sigma(1), \dots, \sigma(M)\}$ --- (*)

Then, clearly $M \geq N$, and for any $i > M$, $|S_i - S'_i| < 2\varepsilon$, so proved.

(More specifically, the (*) is obtained by:

Take $M \in \mathbb{N}$ s.t. $M = \max \{ \sigma^{-1}(1), \dots, \sigma^{-1}(N-1) \}$.

Then, for any $1 \leq i \leq N-1$, $\sigma^{-1}(i) \leq M$, or $\sigma^{-1}(i) \in \{1, \dots, M\}$.

Now, $(\sigma \circ \sigma^{-1})(i) = i \in \sigma[\{1, \dots, M\}]$.

But, if $\sum a_n$ does Not Converge absolutely, the Convergent value cannot be preserved:
in actually, the Convergency can be broken.

Thm. Let $\sum a_n$ be a series of real numbers which converges, but not absolutely.

For any $-\infty \leq \alpha \leq \beta \leq \infty$, there exists a rearrangement $\sum a'_n$ s.t.

$$\liminf_{n \rightarrow \infty} \sum_{k=0}^n a'_k = \alpha \quad \text{and} \quad \limsup_{n \rightarrow \infty} \sum_{k=0}^n a'_k = \beta.$$

Proof. For each $n \geq 1$, let

$$p_n = \frac{|a_n| + a_n}{2}, \quad g_n = \frac{|a_n| - a_n}{2}$$

Then, $p_n + g_n = |a_n|$, $p_n - g_n = a_n$, and $p_n, g_n \geq 0$.

Moreover, both $\sum p_n$ and $\sum g_n$ diverge, because if both are convergent, then

$$\sum_{n=0}^{\infty} |a_n| = \sum_{n=0}^{\infty} p_n + \sum_{n=0}^{\infty} g_n$$

Observe that: for any $n \geq 0$,

converges, it is contradiction.

If only one is convergent, then

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} p_n - \sum_{n=0}^{\infty} g_n$$

$$a_n = \begin{cases} p_n & \text{if } a_n \geq 0 \\ -g_n & \text{if } a_n \leq 0 \end{cases}$$

diverges, and vice versa. Let $\alpha_n = \frac{\alpha}{n}$, $\beta_n = \frac{\beta}{n}$ ($n \geq 2$), $\beta_1 = |\beta|$.

Let m_1, k_1 be the smallest integers s.t.

$$\begin{cases} p_1 + \dots + p_{m_1} > \beta_1 \\ p_1 + \dots + p_{m_1} - (g_1 + \dots + g_{k_1}) < \alpha_1 \end{cases}$$

Let m_2, k_2 be the smallest integers s.t.

$$\begin{cases} p_1 + \dots + p_{m_1} - (g_1 + \dots + g_{k_1}) + p_{m_1+1} + \dots + p_{m_2} > \beta_2 \\ p_1 + \dots + p_{m_1} - (g_1 + \dots + g_{k_1}) + p_{m_1+1} + \dots + p_{m_2} - (g_{k_1+1} + \dots + g_{k_2}) < \alpha_2 \end{cases}$$

Continue in this way, and let x_n, y_n be partial sums as: ($m_0 = 0$)

$$x_n = \sum_{i=1}^{n-1} \left(\sum_{k=m_{i-1}+1}^{m_i} p_k - \sum_{k=m_{i-1}+1}^{m_i} g_k \right) + p_{m_{n-1}+1} + \dots + p_{m_n} > \beta_n$$

$$y_n = \sum_{i=1}^n \left(\sum_{k=m_{i-1}+1}^{m_i} p_k - \sum_{k=m_{i-1}+1}^{m_i} g_k \right) < \alpha_n$$

Then, $|x_n - \beta_n| \leq p_{m_n}$ and $|y_n - \alpha_n| \leq g_{m_n}$. (Because $x_n - p_{m_n} < \beta_n$)